2024 International Conference on Integrated Intelligence and Communication Systems (ICIICS)

Feasible Load Balancing for Webserver in Cloud Environment using Energy Efficient Maximal Support Priority Scheduling Approach

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2024 International Conference on Integrated Intelligence and Communication Systems (ICIICS) | 979-8-3315-0496-0/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICIICS63763.2024.10859387

***Abstract*—Task management and energy efficiency have grown increasingly crucial in cloud computing environments. Effective job scheduling is essential to maximizing resource usage and overall efficiency in cloud computing. However, ensuring energy efficiency is a critical challenge in task scheduling. Traditional task-scheduling methods are often costly and require more adaptability and efficiency. To solve this problem, we propose an Energy Efficient Maximal Support Priority Scheduling Approach (EEMSPS), which schedules tasks in a virtual cloud environment. Furthermore, the Task Completion Time (TCT) algorithm is utilized to calculate the task length and time. Additionally, the Scheduling Based Artificial Bee Colony (SBABC) method is employed to organize the task and reduce data processing time. Lastly, the EEMSPS algorithm assesses the load balancing in the cloud. Additionally, energy efficiency, task execution time, and energy consumption are used to evaluate the proposed method's performance compared to traditional techniques. Therefore, the energy efficiency performance is 96.4% for task scheduling in the cloud.**

Keywords—task scheduling, cloud environment, resource allocation, virtual scheduling, web server, EEMSPS, and SBABC.

I. Introduction

Cloud Data Center (CDC) is an interconnected, virtualized, heterogeneous collection of resources that provides access to computing resources through hosts, storage, applications, and networks. The CDC usually assigns thousands of jobs according to priority. The power consumption of Virtual Machines (VMs) can be dynamically decreased by limiting the number of hosts that can run on them and optimizing resource use [1].

The majority of current energy efficiency devices lose performance to conserve energy. Integrated technology more effectively balances power consumption and host utilization [2]. These resources require minimal organizational tasks for provision and release. A cloud environment combines one or more data centers spread across several geographic locations. Virtualization is the process of transforming physical resources into virtual computing resources. Depending on configuration and availability, user requests are assigned to one or more of these virtual computer resources. The VM Monitor (VMM) or hypervisor components are used to build and manage virtual resources inside the virtual machine [3–4]. Hosts are assigned VMs according to the maximum resources specified in the machine specification.

Many data centers are spread out geographically by cloud service companies such as Google and Amazon. One of the most intricate and energy-intensive technologies is the CDC. CDC can save energy and lower expenses when correctly set up and operated, offering notable system power benefits [5]. Green cloud data centers employ virtualization technologies to increase energy efficiency by integrating VMs [6]. Task scheduling is suggested to move these performed tasks from mobile devices with limited resources to robust VMs [7].

Cloud data center computing systems are not error-free, and increasing their frequency may increase the rapid failure rate and harm reliability. The dependability of computing devices and energy efficiency are competing priorities [8]. However, many data centers need a lot of energy, making cloud computing more expensive [9]. More resources must be used to process requests efficiently, which raises the overhead of energy use and improves the utilization cost of the Physical Machine (PM). Optimizing the allocation of VMs to user-submitted jobs is a difficulty in cloud environments [10]. To consider above problem, this research introduces the EEMSPS method for task scheduling in the cloud.

A. Contribution of this paper

- This paper aims to improve task scheduling based on priority jobs using the proposed EEMSPS method.
- The TCT method is used to identify the task's length and time based matrix.
- Then, SBABC technique, scheduling can be improved, and reduce the data processing time.
- The load balance on a web server in a virtual cloud environment is assessed through performance evaluation.

B. Organization of this research

Section 1 describes the detailed background process, scheduling issues, and contribution of this work. Section 2 defines the related works for scheduling in cloud computing. Section 3 illustrates the proposed methodology for task scheduling in the cloud. Section 4 carries out the result, and finally, the conclusion of this research is represented in section 5.

II. Literature Survey

The author [11] designed a job scheduling technique and implemented a strategy for implementing the proposed taxonomy optimization procedure in cloud computing. However, underutilization or overutilization affects cloud service performance, and improper scheduling is poor in cloud resources. In cloud data centers, tasks are scheduled and allocated based on several heterogeneous Virtual Machines (VMs) to prevent inappropriate scheduling from affecting cloud service performance [12]. However, the data requires the existence of a mechanism provided to reduce the task processing time. The paper presented an Ant Colony

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Optimization (ACO) technique for promptly delivering optimal characteristics by transforming the VM deployment problem into a multi-objective optimization task [13]. However, analyzing the optimal processing method to minimize communication costs is a complex problem. The author developed a new task-scheduling system based on formal adaptation [14]. The Gravitational Search Algorithm (GSA) optimizes the scheduling process's overall time and system VM resource usage. This paper [15] proposed a Load Balancing Algorithm (LBA) to distribute incoming traffic and workload among multiple servers in a cloud computing environment. Furthermore, it provides web services and distributed systems to maintain the stable performance of large-scale systems.

Furthermore [16], the proposed approach improves capacity and availability by maintaining network server performance ratings using a VM clustering process based on bandwidth-level demand workloads. However, VM clusters significantly impact operations due to downtime, complicated backups, and disaster management. Therefore, an ACO scheme based on the neural network method is proposed to address the VM's representation issues [17]. The ACO model is not an optimal method for solving task scheduling problems. However, to address the issue of energy consumption in heterogeneous virtualized clouds and lessen the workload associated with task scheduling, [18] suggested the Energy and Performance Efficient Task Scheduling (EPETS) method. Matching cloud resource power consumption with user needs is a significant challenge. Therefore, to explicitly solve the optimization problem by the cloud environment, a new resource scheduling framework based on the Lyapunov Optimization Approach (LOA) is developed [19]. Furthermore, an optimal trade-off is achieved based on global optimization. Task scheduling has been significantly improved by proposing a Priority-Based Backfilling Backpropagation Neural Network (PBF-NN) technique, which accurately evaluates the makespan and activity costs [20]. However, designing effective workflow planning and cost estimation in a cloud environment is challenging.

TABLE I. CLOUD ENVIRONMENT BASED ON QUEUING APPROACH USING VIRTUAL SCHEDULING

Author	Year	Technique Used	Performance Evaluation	Limitation	Simulator
Ali [21]	2023	Heuristic ACO- Based Multi-Objective Genetic (HACOMOG) algorithm	job execution time, and throughput.	Task allocation is a polynomial time problem, and finding the optimal solution is complex.	CloudSim
Mei [22]	2023	Adaptive Differential Evolution-Simulated Annealing (ADE-SA)	Queue Time	Collaborative decisions regarding formation and scheduling are critical to system design	Lenovo notebook
Malik [23]	2021	Particle Swarm Optimization (PSO)	energy consumption, and load balancing.	Heterogeneous resources, energy efficiency, and resource utilization are some issues cloud computing environments face.	CloudSim
Wang, Y. [24]	2023	Q-learning-based Multi-Task Scheduling Framework (QMTSF)	Makes Pan and Average Task Processing Time.	However, efficient task scheduling remains a significant challenge.	CloudSim
Çavdar [25]	2024	Utilization Based Genetic Algorithm (UBGA)	Networking Bandwidth Consumed.	Selecting the ideal physical machine for hosting the VMs is crucial.	CloudSim

Table 1 indicates the application, performance evaluation, limitations, and simulators of the methods obtained utilizing different serial approach-based virtual scheduling according to the cloud environment.

An Energy and Resource Energy Efficient Workflow Scheduling (ERES) method is presented to dynamically assign VM workflow tasks according to the process task needs of the VM deployment [26]. The Senti Lexicon Demp Score (SLDS)-based word extraction lexicon, which identifies predictively relevant terms [27], is used to process Twitter terms in subject discussions. Furthermore, the proposed EETS method has superior energy consumption accuracy and efficient task scheduling. The author [28] suggested creating a task scheduling framework based on Adaptive Deep Reinforcement Learning (ADRL) to increase the energy efficiency of cloud computing. However, as the energy problem in task scheduling continues to worsen, predicting a fixed number of users per workload is challenging. According to the paper [29], VM scheduling solutions can decrease execution time and increase energy efficiency by examining power usage and the number of instructions performed during the scheduling time. Similarly, the CloudSim toolbox was used to test the Monarch Butterfly Optimization Algorithm (MBO) and assess its performance on synthetic and real-world cloud workloads and optimizations [30].

III. PROPOSED METHODOLOGY

The section describes the proposed EEMSPS method for priority-based load balancing of web servers in cloud environments. The TCT method of cloud computing can compute the load balancing matrix. The SBABC approach can minimize data processing time and enhance scheduling. Performance evaluation assesses web server load balancing in a virtual cloud environment.

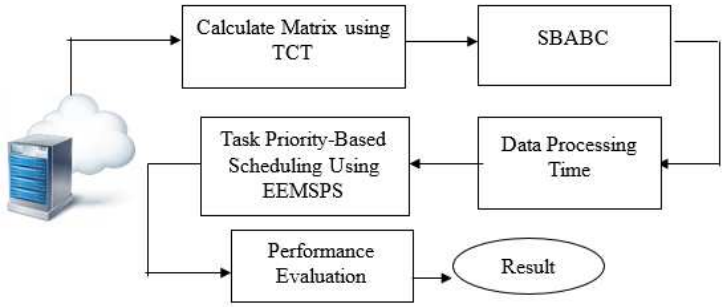

Fig. 1. The Proposed Architecture Diagram Based EEMSPS

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Figure 1 illustrates that the EEMSPS method, based on the proposed architecture diagram, can be utilized for analyzing load balancing in virtual cloud environments and web servers.

A. Task Completion Time (TCT)

This section calculates the matrix for load balancing in cloud computing using the TCT algorithm. Evaluated time in a cloud environment can be used to calculate matrix values given the number of entries available to a VM. Furthermore, by considering different times using cloud computing performance parameters, load balancing can be evaluated based on the variables determining execution time. The execution time in the VM is determined by max pan as the maximum time using the TCT algorithm.

Calculate the processing speed of task length by determining the matrix values shown in Equations 1 and 2. Let's assume O-length, $R_{IA_{ef}}$ −set of task execution time, e-task, f-execution time, S_f −processing speed, y_{ef} −decision variable, and VM −virtual machine.

$$R_{IA_{ef}} = O_e / S_f \tag{1}$$

$$y_{ef} = \begin{cases} 1 \; if \; R_e \in VM_e \\ 2 \; \; if \; R_f \notin VM_f \end{cases} \tag{2}$$

Calculate the VM completion time depending on the decision variables described in Equation 3.

$$R_{if} = \sum_{e=1}^{k} y_{ef} * R_{IA_{ef}} \tag{3}$$

Compute makespan as shown in Equation 4 to calculate the maximum task completion time using a VM. Let's assume M_S −makespan, M-maximum task.

$$M_S = M\left[R_{if}\right]_{f=1}^{K} \tag{4}$$

The TCT algorithm can estimate completion time by determining the load balancing matrix value in cloud computing.

B. Scheduling Based on Artificial Bee Colony (SBABC)

This section discusses methods for improving VM scheduling in load balancing and minimizing data processing times using the SBABC technique for analysis. Cloud environment strategies of server load distribution can be analyzed based on web servers. The SBABC algorithm also provides job scheduling in cloud computing by calculating different load levels for each VM. In addition, the SBABC algorithm can effectively reduce the makespan and response time during load balancing. It calculates the number of bees or the VM's ability to schedule tasks. The number of VMs as processors is counted by allocating the network bandwidth capacity per second to each processor. The most suitable scout bee visitation sites are selected from the vicinity of the VM where the selected bees are located. The standard deviation can be determined by analyzing the workload to measure the deviation of the VM load using the information obtained from the data center.

Analyze the number of bees in the VM to estimate the population's fitness using Equation 5. Let's assume C −capacity, TL −task length, F_{ef} −fitness best population, ef- number of task length.

$$F_{ef} = M_S \frac{\sum_{f=1}^{k} TL_{ef}}{VM_{ef}(C)} \tag{5}$$

As shown in Equation 6, the suitability of sending more bees closer to the ideal VM is estimated. Let's assume IF_L−input file length,

$$F_{ef} = \frac{\sum_{e=1}^{k} TL_{ef} + IF_L}{VM_{ef}(C)} \tag{6}$$

Calculate the average processing time of VMs as shown in Equations 7 and 8. Let's assume y_f −processing time, k − number of VM,

$$y_f = \frac{\sum_{e=1}^{k} TL_{ef}}{C_i} \tag{7}$$

$$\bar{y} = \frac{\sum_{f=1}^{k} y_f}{k} \tag{8}$$

C. Energy Efficient Maximal Support Priority Scheduling (EEMSPS)

This section determines the load balancing on network servers using the proposed EEMSPS algorithm in cloud data center hosts task scheduling. Each server's task sequence can be estimated by analyzing the tasks the cloud data center assigned. Cloud Task Scheduler assigns tasks to server task queues based on global and server-level scheduling data. The EEMSPS algorithm establishes the number of jobs in the queue and the number of tasks allotted to the VM.

Algorithm 1: EEMSPS

Input: processing time← y

Output: Priority-based task allocation ← $R_{I_f^e}$

Start

Determine the number of tasks in the global queue web server.

For each $|U_M|! = 0$ do

Evaluate the task and attributes $Ta_{Type}1$

IF $i = 1,2, \ldots, k$

While $Ta_T P[0] == Ta_{Type}1$

For ← S_{Qi} is full

Continue,

Estimate the completion time ← $R_{I_f^e}$

$S_L.\text{Add} < i, R_{I_f^e} >$

Continue

For each $S_L! = \phi$

Evaluate the minimum time ← $R_{I_f^e}$

Assign Ta_j to S_i

Remove task Ta_j ← U_M

Determine the period of each task assigned to the server.

$R_{I_f^e} = (W_{Pi} + W_{Di} + 1)A_T$

End for each

End if

End while

End for

End for each ← $R_{I_f^e}$

Return

End

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Let's assume Ta_T −task type, P-priority, S_{Qi} −server task queue, T − time, S_L, ϕ − server list, A_T − average time, W_{Pi} −weighting Priority, U_M −global queue.

IV. Result And Discussion

This section evaluates the performance of test results predictions using the proposed EEMSPS algorithm. Furthermore, performance metrics such as task execution time, energy consumption, Degree of Imbalance (DOI), and energy efficiency can be compared with the offered method. Similarly, the performance evaluates the suggested EEMSPS technique for formulation and experimental simulation parameters. Calculate resource utilization metrics in cloud environments while scheduling VMs to maximize energy efficiency within virtual scheduling. Moreover, the results show that the execution time of the proposed model is lower than that of other techniques by reducing the scheduling cost of the VMs. In addition, execution efficiency is defined by analyzing the load distribution level in the VM and is expressed as the minimum DOI value. Furthermore, the proposed EEMSPS method can be compared with the previous GSA, EPETS, and HACOMOG methods to provide experimental results based on performance evaluation.

TABLE II. Simulation Parameter

Simulation	Variable
Language	C#
Tool	Visual Studio
Cloud environment	Amazon Web Service (AWS)
Number of Datacenter	10
Bandwidth	2000
Total number of VMs	40-220
Virtual Machine Memory (MB)	512-2049
Total number of Task	150-750
Task Length	5500-20000
Cost of memory	0.5
Cost of execution	1.0
Number of processing elements	1
Speed Processing Element	10000MIPs

Table 2 shows simulation parameters that can be defined for AWS web server resource scheduling in a cloud environment using Visual Studio tools based on C# language.

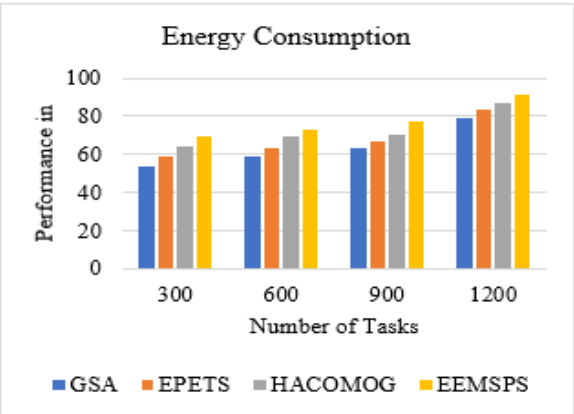

Fig. 2. Analysis of Energy Consumption

As illustrated in Figure 2, we offer load balancing and energy-saving resource scheduling for web servers in cloud environments based on the energy consumption analysis. Furthermore, other GSA, EPETS, and HACOMOG methods can be compared with the proposed EEMSPS method to estimate the number of tasks for energy consumption analysis. Thus, the offered EEMSPS approach increases the efficiency of VM scheduling and allocation in cloud data centers by 91%. However, when evaluating other methods in the analysis of energy consumption, the proportions are 79%, 83%, and 87%.

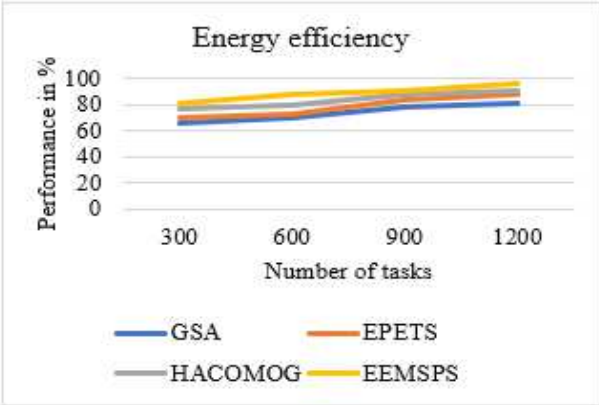

Fig. 3. Analysis of Energy Efficiency

According to the energy efficiency analysis, we offer resource scheduling and feasible load balancing for web servers in cloud environments, as illustrated in Figure 3. Other GSA EPETS and HACOMOG methods can be compared with the proposed EEMSPS method to evaluate the number of tasks for energy efficiency analysis. As a result, the presented EEMSPS approach enhances the efficiency of VM scheduling and allocation in cloud data centers by 96.4%. However, when evaluating other methods in energy efficiency analysis, the percentages are 80.6%, 87%, and 91%.

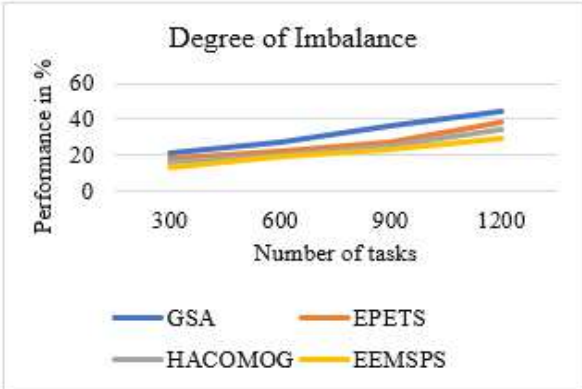

Fig. 4. Analysis of Degree of Imbalance

The level of load distribution among VMs can be defined by implementation capabilities, providing resource scheduling and feasible load balancing for web servers based on the minimum range of DoI, as shown in Figure 4. In addition, comparing the previous GSA, EPETS and HACOMOG methods with the proposed EEMSPS approach can help estimate the number of tasks for the DoI analysis. The proposed EEMSPS approach demonstrates an efficiency of 29.4% for VM scheduling and allocation in cloud data centers. Similarly, the efficiency ratios were 49.8%, 41.2%, and 34% when evaluating other methods in DoI analysis.

Job execution time analysis may be assessed by offering resource scheduling and potential load balancing for web servers in a cloud context, as seen in Figure 5. Moreover, the number of processes employed for task execution time analysis will be estimated by contrasting the suggested EEMSPS method with earlier GSA EPETS and HACOMOG

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methods. The proposed EEMSPS method for allocating and scheduling VMs in cloud data centers has an efficiency of 18.5 ms. Comparably, while assessing alternative approaches in task execution time analysis, the efficiency ratios were 27 ms, 24 ms, and 21 ms.

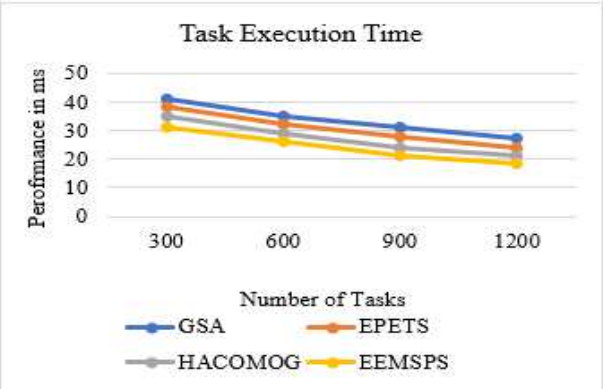

Fig. 5. Analysis of Task Execution Time

V. Conclusion

In conclusion, the load balancing of web servers and virtual cloud environments may be examined using the EEMSPS approach based on the suggested architecture design. Additionally, the TCT algorithm can calculate the load balance matrix in cloud computing. Then, VM scheduling can be improved by implementing the SBABC method to reduce data processing time. Additionally, energy efficiency, job execution time, imbalance degree, and energy consumption can be used to assess how well the suggested EEMSPS approach performs compared to the prior art. The recommended approach greatly increases energy efficiency over earlier GSA, EPETS, and HACOMOG methods. Furthermore, by analyzing the performance evaluation, the energy efficiency of the proposed method was improved to 96.4%. However, this work didn't find the on-demand task request. Therefore, in the future, we will introduce the priority based on the on-demand request rate and arrival rate for scheduling in the cloud.

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